Chemicals of Security Concern (CoSC) Risk Reduction Goals

Technical Solution Performance Goals: The performance criteria of system size, reactant ratio (reactive candidates), and scalability are applicable to every case; however, any combination of the other three performance criteria (reduction in toxicity, reduction of the ability to disseminate , reduction in volatility) that achieves a 50% reduction in overall hazard, whether through a single criterion or combination, is acceptable.

Performance Goals. The following are the criteria for technical solution performance:

1. **System Size:**
   - **Definitions**
     - Large Bulk Liquid Containers: 1000L totes.
     - Large Bulk Gas Containers: 100lb compressed gas cylinders.
   - **Restrictions**
     - If the proposed solution is readily available at a Chemical Production Facility, this criterion will not apply.
     - System footprint is not restricted
   - **Goals**
     - The proposed technical solution's mass is:
       1. Threshold: Less than the mass of the target CoSC.
          1. Preferred: ≤ 500 kg.
          2. Ceiling Value: Large bulk container solutions have a maximum mass (ceiling weight of equipment) of 3,000kg.
       2. Objective: ≤ 36 kg.

2. **Reactant Ratio:**
   - **Definitions**
     - Reactant: A substance added to the CoSC to alter its chemical properties.
   - **Restrictions**
     - If the technical solution does not require the addition of a reactant, this criterion will not apply.
   - **Goals**
     - If reactants are used, the reactant-to-CoSC ratio is:
       1. Threshold: ≤ 2:1.
       2. Objective: <1:1.

3. **Indication/Proof of Scalability:**
   - **Definitions**
     - Sub-scale: Less than 100mL (minimum required for analytical laboratory testing).
     - Laboratory Scale: 100-1000mL.
     - Field-Testing Scale: 1-10L.
     - Bulk Container Scale: 1000L totes (liquid) and 100lb compressed gas cylinders (gas).
   - **Restrictions**
     - N/A
   - **Goals**
     - The approach meets, or is expected to meet, the applicable performance goals at four scales: subscale, laboratory scale, field-testing scale, and bulk container scale.

Reducing Toxicity (Evaluated Only if the Solution Does Not Focus on Disseminability)

1. **Reduction in Toxicity**
   - **Definitions**
     - Toxicity: Measured based on the concentration of the CoSC present and based on the following exposure limts.
       1. IDLH – immediately dangerous to life and health
       2. PEL -the OSHA Permissible Exposure Limit. If the PEL is not published, another federal exposure limits may be used.
   - **Restrictions**
     - Set-up time is not included in the duration evaluation.
     - The toxicity of the resulting reactants will only be considered if their exposure limits are equal to or higher than the original CoSC.
   - **Goals**
     - The technical solution will reduce the CoSC's toxicity:
       1. Threshold: By 50%.
       2. Objective: below the PEL.

       AND

     - The reduction in toxicity will occur within:
       1. Threshold: ≤ 7 days.
       2. Objective: ≤ 2 hours.

Performance Goals for Reducing Disseminability (Evaluated Only if the Solution Does Not Focus on Toxicity)

To pass the performance goals for reducing disseminability, only one of the following criteria must be met.

1. **Reduction in Disseminability**
   - **Definitions**
     - Disseminability -the potential to be aerosolized and/or weaponized/ released for harmful purposes.
   - **Restrictions**
     - Evaluation of a solution's disseminability will be measured qualitatively to assess its weaponization potential using CBARR's subject matter expertise.
   - **Goals**
     - The technical solution will reduce the CoSC's disseminability by:
       1. Threshold: Transforming its final state into a non-aersolizable liquid or gel.
       2. Objective: Transforming its final state into a solid.

2. **Reduction in Vapor Pressure**
   - **Definitions**
     - Vapor pressure will be measured using a standard method or evaluated based on published values.
   - **Restrictions**
     - N/A
   - **Goals**

- The technical solution will reduce the CoSC's vapor pressure by:
  1. Threshold: 50% reduction from the initial value.
     a. Alternative Approach: Sealing the surface of the CoSC to prevent vapor escape will meet the threshold.
  2. Objective: Achieving a vapor pressure of $\leq$0.0007 mmHg.

3. **Reduction in Volatility**
   - **Definitions**
     - Volatility will be measured at 25°C.
   - **Restrictions**
     - N/A
   - **Goals**
     - The technical solution will reduce the CoSC's volatility by:
       1. Threshold: 50% reduction from the initial value.
          a. Alternative Approach: Sealing the surface/encapsulation of the CoSC to prevent vapor escape will meet the threshold.
       2. Objective: Achieving a volatility of $\leq$ 10.5 mg/m^3.